Multiscale and Cross-Level Attention Learning for Hyperspectral Image Classification

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***Abstract*— Transformer-based networks, which can well model the global characteristics of inputted data using the attention mechanism, have been widely applied to hyperspectral image (HSI) classification and achieved promising results. However, the existing networks fail to explore complex local land cover structures in different scales of shapes in hyperspectral remote sensing images. Therefore, a novel network named multiscale and cross-level attention learning (MCAL) network is proposed to fully explore both the global and local multiscale features of pixels for classification. To encounter local spatial context of pixels in the transformer, a multiscale feature extraction (MSFE) module is constructed and implemented into the transformer-based networks. Moreover, a cross-level feature fusion (CLFF) module is proposed to adaptively fuse features from the hierarchical structure of MSFEs using the attention mechanism. Finally, the spectral attention module (SAM) is implemented prior to the hierarchical structure of MSFEs, by which both the spatial context and spectral information are jointly emphasized for hyperspectral classification. Experiments over several benchmark datasets demonstrate that the proposed MCAL obviously outperforms both the convolutional neural network (CNN)-based and transformer-based state-of-the-art networks for hyperspectral classification.**

***Index Terms*— Hyperspectral image (HSI) classification, multihead self-attention (MHSA), multiscale convolution (MSC), transformer.**

I. Introduction

HYPERSPECTRAL remote sensing technology is a multidimensional information acquisition technology that combines imaging technology and spectral technology. Compared with ordinary RGB images, hyperspectral images (HSIs) own higher spectral resolution, more bands, and thus each pixel has an independent spectral curve, which helps HSI to detect the physical features of materials [1], [2], [3], [4]. HSI classification refers to identifying the category of each pixel in an image and has been widely used in environmental monitoring [5], geological exploration [6], intelligent agriculture [7], and other fields.

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In the early stages of research, hyperspectral classification mainly focused on the spectral features, including kernel-based support vector machines (SVMs) [8], principal component analysis (PCA) [9], k-nearest neighbors (KNN) [10], and so on. However, these methods often ignore the spatial information in HSIs. To better extract the spatial features, the researchers proposed to extract the spatial–spectral features jointly by the wavelet filter [11] and Gabor filter [12]. Recently, some innovative methods such as the sparse representation (SR)-based graph embedding method [13], [14] and scheduling-guided method [15] have been extensively used in the HSI feature extraction tasks. Considering the multivariate sparse manifold correlation between samples, Duan et al. [13] constructed a pair of semi-supervised hypergraphs, which effectively fused labeled and unlabeled training information into the embedding process to obtain the nonlinear spectral–spatial features of HSI. Wu et al. [15] proposed a new divisible scheduling model to make full use of the intertask and intratask parallelism of the representative HSI classification method in distributed processing. But these conventional classifications have obvious drawbacks. On one hand, these methods rely too much on prior information and are time-consuming and labor-intensive. On the other hand, they are difficult to mine deep nonlinear spectral–spatial correlation information.

Compared with the conventional classifiers, deep learning (DL)-based methods have been rapidly developed in visual tasks due to their powerful fitting ability and feature extraction ability. Recently, the DL-based classifiers automatically discover the discriminative representations of interest and have been widely used in HSI classification. Representative DL networks include stacked autoencoders (SAEs) [16], deep belief networks (DBNs) [17], recurrent neural networks (RNNs) [18], and convolutional neural networks (CNNs) [19], [20], [21], [22], [23]. The CNN architecture captures the local information using local receptive fields, shared parameters, and spatial subsampling, while being able to skillfully combine the spectral features with spatial context information in HSI data. Therefore, in the above network, CNN has gradually become a research hotspot in HSI classification. The CNN mainly includes 1-D-CNN, 2-D-CNN, 3-D-CNN, and their variants. Among them, 1-D-CNN [24], [25] is able to extract vectorized features along the spectral dimension. The 2-D-CNN [26], [27], [28], [29], [30] is able to extract features along the spatial dimension. The 3-D-CNN [31], [32], [33], [34], [35] handles both the spectral and spatial domains using 3-D convolutions.

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The extracted features not only contain spectral information and spatial information but also the fusion information of the two. Due to the characteristics of rich spectral channels and strong spatial correlation in HSIs, 3-D-CNN is more suitable for the processing of HSIs.

However, there are still many deficiencies in the structure of CNNs. First, the features extracted by CNN usually contain noise as interference to affect the classification results. Especially in the case of small samples, the classification accuracy will be greatly affected. Second, due to kernel limitations, the receptive fields of the CNN-based methods are limited to the squared region of the convolution kernel size, which usually makes the network unable to model long-range dependencies. In this case, the researchers designed a series of units to enhance the feature extraction ability of the network, and the attention mechanism [36], [37], [38], [39], [40], [41], [42], [43], [44], [45], [46] is one of the enhanced units that have been intensively studied in recent years. The spectral–spatial attention mechanism proposed by Sun et al. [37] can suppress the influence of the interfering pixels and extract more discriminative spectral–spatial features. Zheng et al. [39] focused on solving the problem of small receptive field of CNN and proposed a global attention mechanism. The method directly processes the entire image through the encoder and the decoder and adopts a global stochastic layering strategy to ensure fast processing and depth of the image. Although the CNN with the attention mechanism module has achieved good results in HSI classification, it still cannot get rid of the limitation of the backbone network structure (CNN). The correlation between different spectral bands and spatial pixels and the sequence properties between spectral bands cannot be completely described by simple attention models.

Recently, a novel architecture, named transformer, has attracted more and more interests, which is originally proposed for machine translation in natural language processing (NLP). Transformer architecture is a self-concerned-based structure that combines multilayer perceptrons (MLPs) to model the long-distance correlation of tokens in a sequence. With the introduction of the vision transformer (VIT) [47] and its variants, the transformer-based networks have approached or even surpassed the CNN-based networks. The biggest advantage of the transformer is that it can capture global information about images. It is this advantage that it can be well-applied to the classification task of HSI with hundreds of continuous spectral bands [48], [49], [50], [51], [52], [53], [54], [55], [56]. However, the transformer-based models still suffer from some problems in dealing with the HSI classification task. First, the traditional transformers model the contextual dependencies of input patches, but lack the modeling of local information with rich spatial patterns and geometric structures. Second, although the transformer captures global information by building correlations between tokens, it cannot fully consider the multiscale information of a single sample.

To address the above problems, a novel network named multiscale and cross-level attention learning (MCAL) network is proposed, which includes a spectral attention module (SAM), a multiscale feature extraction (MSFE) module, and a cross-level feature fusion (CLFF) module. The network combines convolution with the multihead self-attention (MHSA) to extract multiscale local features and global features of pixels. And to model the correlation between shallow features (shape, texture) and deep features (semantic information) in HSI, the network adopts a hierarchical structure and can adaptively fuse features from the hierarchical structure of MSFEs using the attention mechanism. Finally, experiments over several benchmark datasets, including Indian Pines (IP), Pavia Center (PC), HyRANK-Loukia (HR-L), and WHU-HanChuan (WHU-HC), are conducted to verify the effectiveness of the proposed MCAL for classification of HSIs.

The main contributions are as follows.

1) An MCAL network is proposed for hyperspectral classification by enhancing the multiscale local features in the transformer-based networks, by which both the global and local multiscale features are integrated for classification of hyperspectral pixels.
2) An MSFE module is constructed to explore both the multiscale local spatial feature of pixels using different sizes of convolutions and the global features of pixels using MHSA.
3) A CLFF module is proposed to emphasize different levels of deep features extracted by hierarchical stacking of MSFEs using the attention mechanism.
4) The spectral information is emphasized for hyperspectral classification using SAM prior to hierarchical stacking of MSFEs.

The remaining sections of this article are organized as follows. First, Section II briefly reviews some recent works related to HSI classification. Second, Section III describes the details of our proposed method. Third, in Section IV, a series of experiments and results are illustrated to compare our model with several sate-of-the-art methods. Finally, some conclusions of this article are drawn in Section V.

II. Related Work

A. Conventional CNN-Based Methods

When CNNs are used for HSI classification, they mainly rely on the feature extraction networks within CNNs. The network can directly process 3-D HSI blocks, avoid complex preprocessing of images, and effectively extract spectral and spatial background information, which greatly improves the accuracy of HSI classification compared with the traditional methods. According to the extracted feature dimensions, the CNN models can be divided into 1-DCNN, 2-DCNN, and 3-DCNN.

In the early classification stage of HSI, 1-DCNN is used to extract vectorized features along the spectral dimension. Hu et al. [24] explored how to apply CNNs to HSI classification and proposed a CNN with only one convolutional layer and one fully connected (FC) layer. Yue et al. [25] introduced 1-DCNN to HSI classification and extracted spectral features on the basis of using the PCA algorithm to reduce spectral information redundancy. However, 1-D-CNN ignores the rich spatial information contained in HSIs.

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Compared with 1-D-CNN, 2-D-CNN can handle the spatial dimension of HSI and effectively capture the spatial information required for classification tasks. Mei et al. [26] proposed a five-layer classification framework based on CNN, which integrated the spectral features and spatial context into the framework by preprocessing adjacent pixels, while fully considering their unique characteristics. Using CNN and MLP, Makantasis et al. [27] proposed a model that automatically captures high-level spatial features for HSI classification. Mei et al. [28] proposed a peer feature learning CNN based on supervised and unsupervised modes to learn sensor-specific spatial–spectral features. Chen et al. [29] proposed a network model combining the Gabor filter with CNN, which can effectively extract the spatial features including edges and textures while preventing overfitting. Liu et al. [30] extracted and aggregated local and global spatial features through 2-D-CNN, while using more representative class features instead of redundant pixel information in the reconstruction process.

However, studies have shown that considering spectral information and spatial information in isolation cannot handle the complex 3-D structure of HSI very well, and often ignores the joint features of space and spectrum, which will have a serious impact on HSI classification that needs to be accurate to each pixel category. Therefore, 3-DCNN is proposed to fuse information from 2-D. Using 3-D convolution kernels to jointly slide between the spatial and spectral dimensions, 3-D-CNN can model the 3-D structure of HSI in a more natural way. Zhong et al. [31] proposed a 3-D convolution-based network model that extracts the spatial–spectral features using cascaded residual blocks. Hamida et al. [32] proposed a new 3-D method to realize the joint processing of the spatial and spectral domains, which not only improves the classification accuracy but also provides in-depth image understanding at a lower cost. Roy et al. [33] proposed a hybrid model based on 2-DCNN and 3-DCNN, where 3-DCNN is used to learn the spatial–spectral joint features, and 2-DCNN is used to further learn the spatial features. Zheng et al. [34] added the covariance pooling technology and PCA analysis strategy based on the mixed model of 2-DCNN and 3-DCNN. This method reduces the amount of parameters that need to be calculated and also improves the classification accuracy. Liang et al. [35] proposed a lightweight DL framework based on the goal-directed fusion mechanism to enhance the correction effect of adjacent spectra when fusing multiscale features.

B. CNN-Based Methods With Attention Mechanism

The attention mechanism is applied in the CNN architecture as an enhancement unit. On one hand, the introduction of the attention mechanism can enhance the extraction of useful information, suppress useless information, and improve the classification results in the case of small samples. On the other hand, it can solve the influence of CNN receptive field limited by the size of convolution kernel by calculating the dynamic global weights.

Sun et al. [37] proposed a spectral–spatial attention network, which can acquire discriminative spectral–spatial features while suppressing the influence of the interfering pixels. Lu et al. [38] enhanced the expressiveness of the spatial–spectral features using a parallel residual network structure to extract the deep features of HSI and input them into a 3-D attention module. Zheng et al. [39] designed a fast patch-free global learning framework for HSI classification. The framework directly processes the entire image through the encoder and the decoder and adopts a global random layering strategy to ensure fast processing and depth of the image. Hang et al. [40] proposed a spectral attention subnetwork and a spatial attention subnetwork for HSI classification, which can help the network focus on more discriminative spectral information or location information. Zhao et al. [41] designed a central attention module, which focuses on the central pixels in the sample, and then weights various features to find more relevant and discriminative features. Tang et al. [42] proposed a 3-D octave convolution with the spatial–spectral attention network, in which two attention methods are introduced to capture the significant spatial areas and explore the specific spectral bands. Zhu et al. [43] combined the SAM and spatial attention module with the residual network structure. The designed network can not only emphasize useful bands and suppress heterogeneous pixels but also avoid overfitting and speed up the image processing speed. Shi et al. [44] incorporated a multibranch hybrid CNN with multiple resolutions and attention mechanism to emphasize useful information while extracting multiscale features. Mei et al. [45] fully explores the bidirectional relationship between adjacent band images in an HSI by taking the neighborhood block of each pixel as the input of the attention network and passing through the location attention network and channel attention network in turn. He et al. [46] proposed a dual attention module to complete the correction of the original data by grasping the spatial and spectral features from a global and local perspective.

C. VIT-Based Methods

Transformers have achieved great success in various fields in recent years. It is completely different from CNN, which is a structure based on the attention mechanism combined with MLP, and can capture global information. The proposal of VIT enables the transformer to be successfully applied to the image domain and has a performance equal to or even better than that of CNN.

Recently, many researchers have introduced transformers to HSI classification and derived many transformer-based models. Hong et al. [48] proposed a transformer network architecture that focuses more on spectral learning and is combined with cross-layer connectivity. Qing et al. [49] designed an end-to-end model based on a transformer, which simultaneously introduced an SAM so that it could introduce a learnable embedding vector during the encoding process. He et al. [50] cascaded CNN and transformer, where CNN is used to extract the spatial features, while transformer is used to focus more on spectral feature extraction. Tan et al. [51] sent the extended morphological profile (EMP) features of HSI into the transformer and extracted the patch-level and pixel-level features, respectively, to make full use of the global and local information of the EMP. He et al. [52] used bidirectional

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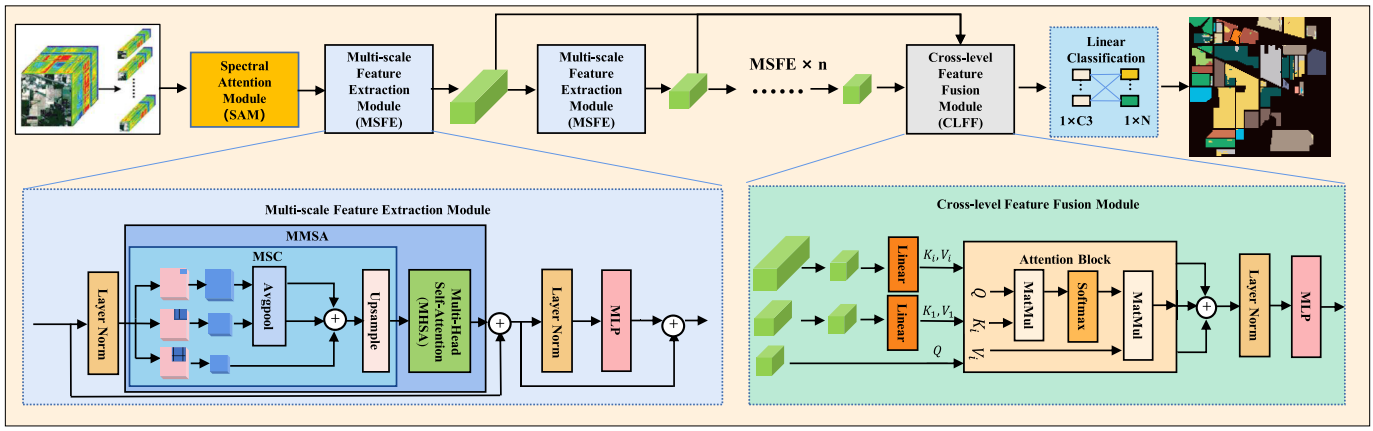


Fig. 1. General schematic of the proposed network model.

encoder representation from transformers to extract features from HSI, which is more flexible than CNN and has a larger receptive field and stronger generalization ability. Sun et al. [53] also cascaded CNN and transformer and added a Gaussian weighted feature marker machine between the two to further improve the classification effect. Xue et al. [54] introduced a new attention mechanism in the transformer, which relies on sub-blocks obtained by sliding around the central pixel point. Liu et al. [55] proposed a central attention mechanism to extract the spectral–spatial information by exploiting a central pixel and pixels similar to the central pixel on an HSI patch. Mei et al. [56] designed a new and efficient block pixel embedding module that is able to model local components as a complement to the overseparated features extracted from transformers.

III. Proposed Method

A. Overall Framework

The framework of our proposed MCAL is illustrated in Fig. 1, which consists of an SAM, an MSFE, a CLFF module, and a linear classification module. First, to suppress redundant information in spectral bands, the SAM is adopted to adaptively recalibrate spectral bands by enhancing information bands and weakening useless bands. Second, to extract the global information and local information in HSI while taking into account the multiscale information, the MSFE is proposed to perform feature extraction on SAM-processed HSI by combining the CNN and transformer. Moreover, MSFE is stacked several times to build a hierarchy that reduces the number of channels of feature maps in a hierarchical manner to model low-level, mid-level, and high-level features for classification. Third, to aggregate the features extracted by MSFE at different levels, the CLFF is proposed, which uses an attention mechanism to model the correlation of features at different levels. Finally, the extracted depth features are linearized into a 1-D vector to complete the classification The above modules will be described in detail.

B. Spectral Attention Module

The SAM in residual spectral–spatial attention network (RSSAN) [43] is introduced into the proposed MCAL network,

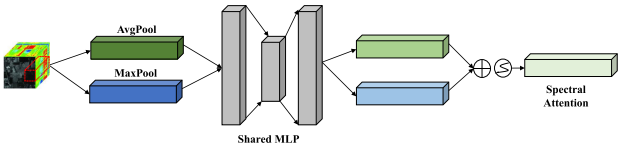


Fig. 2. Structure of SAM.

aiming to discover the bands that are important for HSI classification and prepare for subsequent image processing. The structure of SAM is shown in Fig. 2.

Given an input $X \in R^{H\times W\times C}$, after SAM processing, a 1-D spectral attention map $M_{\text{se}} \in R^{1\times 1\times C}$ will be generated. Perform elementwise multiplication of this attention map with the input to complete the processing of spectral bands. The above attention calculation process is summarized as follows.

First, the input $X \in R^{H\times W\times C}$ is processed by a global average pooling layer and a global max pooling layer, respectively. Let z represent the features processed by the pooling layer, and the cth element of z_c^{avg} and $z_c^{\max}$ is calculated by the following equation:

$$z_c^{\text{avg}} = \frac{1}{H\times W}\sum_{i=1}^{H}\sum_{j=1}^{W} x_c(i,j) \tag{1}$$

$$z_c^{\max} = \text{Max}(x_c) \tag{2}$$

where $x_c(i,j)$ is the value at position (i,j) of the cth channel. It then goes through two FC layers, i.e., the MLP in Fig. 2. Among the MLP, the first FC layer is a dimensionality-reduction layer parameterized by W_1, and the second is a dimensionality-increasing layer parameterized by W_2. To reduce the number of parameters, the parameters $\{W_1, W_2\}$ of two FC layers are shared for the output of both global average pooling and global max pooling. This part operates as follows:

$$s_c^{\text{avg}} = \text{Relu}\big(W_2\big(\text{Sigmoid}\big(W_1 z_c^{\text{avg}}\big)\big)\big) \tag{3}$$

$$s_c^{\max} = \text{Relu}\big(W_2\big(\text{Sigmoid}\big(W_1 z_c^{\max}\big)\big)\big). \tag{4}$$

Then, the outputs of two FC layers s_c^{avg} and s_c^{avg} are added together

$$s_c = s_c^{\text{avg}} + s_c^{\max}. \tag{5}$$

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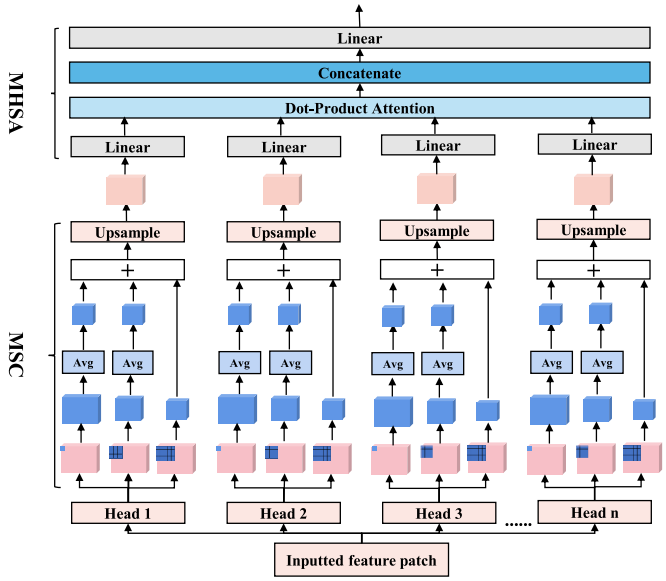


Fig. 3. Structure of MMSA.

Concatenate the weights of each spectral band to get M_{se}

$$M_{\text{se}} = \text{Concat}(s_1, s_2, \ldots, s_c, \ldots, s_n) \tag{6}$$

where Concat$(\cdot)$ represents the connection of each channel. Finally, perform the elementwise multiplication with the input X to complete the processing of the spectral dimension

$$X' = M_{\text{se}}(X) \otimes X. \tag{7}$$

C. MSFE Module

Transformer has shown its advantages in building global long-range dependencies and extracting important features, but it is insufficient in the processing of HSI local multiscale information. To make up for the inherent defect of transformer and fully understand the information of HSIs, the MSFE module is proposed to extract multiscale features by embedding multiscale convolution (MSC) inside the transformer. The extracted features not only have global information but also take into account local multiscale information. The structure of the MSFE module is described in detail below.

As shown in Fig. 1, MSFE consists of multiscale MHSA (MMSA) and MLP. Meanwhile, the layer-norm (LN) layer is used before each block, and the residual connection is used after each block. Similar to the original transformer, the MLP block consists of two FC layers with a ReLU activation function. The core part of MSFE is MMSA, which consists of MSC and MHSA. The structure is shown in Fig. 3. MSC consists of convolutional layers, average pooling layers, and upsampling layers, which aims to complete multiscale information extraction from HSIs through convolution kernels of different sizes. MHSA consists of a linear layer, a dot-product attention layer, and a concatenate layer, which aims to model the sequence properties of HSIs and establish global long-range dependencies. Below, the MMSA is described in detail.

Unlike conventional transformers, the input patches are not reshaped into a 2-D form, but directly inputted into a 3-D form. The reason is that maintaining the invariant structure of the original image patch can maintain the spatial structure of HSI and is conducive to extracting local multiscale information.

For input X, it is first divided into n groups along the spectral dimension

$$X = \{X_1, X_2, \ldots, X_i, \ldots, X_n\} \tag{8}$$

where n is the number of groups, and i is the serial number of the group. Each group goes through a layer of convolution operation that contains three convolution kernels of different scales. The sizes of the convolution kernel are 1×1, 3×3, and 5×5, respectively. The process can be formulated as follows:

$$X_{i,k} = \text{C}_{\text{k}}(X_i) \tag{9}$$

where $X_{i,k}$ represents the feature generated by the kth (k = 1,2,3) convolutional layer C_{k} in Fig. 3. Thus, multiscale information is extracted under different spatial receptive fields in HSI by adopting three convolution kernels of different sizes. The existing methods show that using the combination of the convolution layer and global average pooling layer to replace the combination of the convolution layer and full connection layer is equivalent to regularization in the network structure to prevent the overfitting of the model. Hence, the output feature maps of different sizes are downsampled to the smallest feature map size through a global average pooling layer

$$X'_{i,k} = \text{Avg}\left(X_{i,k}\right) \tag{10}$$

where Avg$(\cdot)$ represents the global average pooling layer. Add these features together to obtain fusion features, which contain multiscale information

$$F_i = \sum_{k=1}^{3} X'_{i,k}. \tag{11}$$

Then, the fused feature map is upsampled by deconvolution to make the size of the feature map consistent with the input patch size. In this way, the fuzzy feature of nodes caused by the small feature map can be alleviated, and the ability of the network to collect feature information can be further enhanced.

Next, the fusion features are converted into a 2-D sequence form and inputted into MHSA. The input 2-D sequence $F \in R^{s \times d}$ is mapped to three different matrices including query (Q), key (K), and value (V) by linear transformations. The formula is as follows:

$$Q_i = \text{FW}_i^Q, \quad K_i = \text{FW}_i^K, \quad V_i = \text{FW}_i^V \tag{12}$$

where s and d are the length and dimension of the input sequence, respectively, and $W_i^Q, W_i^K, W_i^V \in R^{d \times d}$ are the learned parameter matrices. The output of each header is computed as a weighted sum of the values, where the weight assigned to each value is computed by a compatibility function Softmax of the query with the corresponding key. The formula is as follows:

$$Z_i = \text{Attention}(Q_i, K_i, V_i) = \text{Softmax}(Q_i, K_i)V_i. \tag{13}$$

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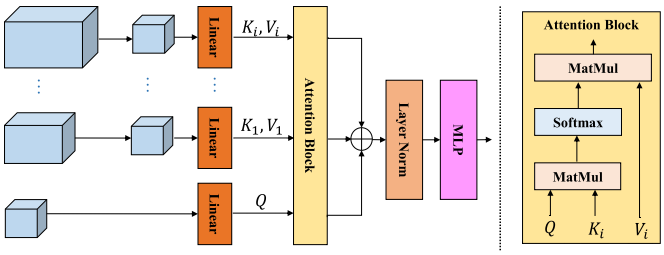


Fig. 4. Structure of CLFF.

The final output of MSFE is generated by a projection after concatenating all $Z_i \in R^{s\times(d/n)}$, which can be calculated as follows:

$$Z_{\text{Out}} = \text{Concat}\{Z_1, Z_2, \ldots, Z_n\}W^O \tag{14}$$

where W^O represents the output projection matrix.

The above MSFE is stacked several times to build the hierarchy and halve the spectral dimensions one by one, aiming to model both shallow and deep features for classification. This is because shallow features often contain low-level information of images such as shape and texture, while deep features often contain semantic information of images. Considering the complexity and diversity of terrain distribution, features from different levels contribute to HSI scene classification.

D. CLFF Module

As mentioned before, due to the complexity of HSIs, both shallow and deep features contribute to the performance of the classification task. However, if only shallow features and deep features are entered into the classifier separately, the correlation between them will be ignored. At the same time, the internal features of an object are not simply stacked together, but closely related in some relationship. To build this correlation and aggregate multilevel features, the CLFF module is proposed. The structure is shown in Fig. 4.

Taking the network with three MSFE modules as an example, the features from three different levels are, respectively, called low-level features, mid-level features, and high-level features. First, the output feature size of each level of MSFE is unified, and the low-level features $Z_L \in R^{H\times W\times C_L}$ and mid-level features $Z_M \in R^{H\times W\times C_M}$ are transformed into the same size as the high-level $Z_H \in R^{H\times W\times C_H}$ features using the linear layer. The new features are $Z'_L \in R^{H\times W\times C_H}$ and $Z'_M \in R^{H\times W\times C_H}$

$$Z'_L = \text{Linear}(Z_L) \tag{15}$$
$$Z'_M = \text{Linear}(Z_M) \tag{16}$$

where Linear represents the linear layer. Afterward, two learnable parameters W^K and W^V are used to project Z'_L and Z'_M into the spaces of K and V. One learnable parameters W^Q are used to project Z_H into the spaces of Q. The above steps can be expressed as follows:

$$Q = Z_H W^Q \tag{17}$$
$$K_1 = Z'_L W^K, \quad V_1 = Z'_L W^V \tag{18}$$
$$K_2 = Z'_M W^K, \quad V_2 = Z'_M W^V. \tag{19}$$

Next, the correlation weights between Q and K are calculated using the dot product and softmax function, and then multiplied by V to generate aggregate features

$$Y_1 = \text{Softmax}(Q, K_1)V_1 \tag{20}$$
$$Y_2 = \text{Softmax}(Q, K_2)V_2 \tag{21}$$
$$Y = Y_1 + Y_2 \tag{22}$$

where Y represents the aggregate features.

Finally, the features are successively passed through LN and MLP and combined with residual connection to prevent overfitting problem.

IV. Experiments

A. Datasets

To evaluate the performance of our proposed network model, experiments over four benchmark datasets are conducted, including IP, PC, HR-L, and WHU-HC.

1) Indian Pines: The IP dataset was imaged by the airborne visible infrared imaging spectrometer (AVIRIS) in 1992 of an Indian pine tree in Indiana, USA. The original image has 220 bands with wavelengths ranging from 0.4 to 2.5 μm. After removing the water absorption bands, the remaining 200 bands were used finally as study objects. The data size is 145 × 145. There are 10 249 ground object pixels in total, representing 16 ground object categories.

2) Pavia Center: The PC dataset was obtained by the ROSIS sensor in 1996. The original image has 115 bands with wavelengths ranging from 380 to 860 nm. After removing the water absorption bands, 102 bands were used for evaluating. The height and width of the data are 1096 and 715 pixels, respectively. The dataset has 148 152 marked samples presenting nine object classes.

3) HyRANK-Loukia: The HR-L dataset is one of the five satellite HSIs from the EO-1 Hyperion sensor in the framework of the International Society for Photogrammetry and Remote Sensing (ISPRS) scientific initiatives. The HR-L image consists of 249 × 945 pixels and 176 spectral bands. In this dataset, a total number of 13 503 pixels are manually labeled and 14 different land covers are included.

4) WHU-Hi-HanChuan (WHU-HC): The WHU-HC dataset was collected by the Headwall Nano-Hyperspec sensor in Hanchuan City, Hubei Province, China. The spectral imager continuously images 274 bands in the wavelength range of 400–1000 nm. The spatial resolution of the image is 0.109 m. The size of the data is 1217 × 303, and there are 257 530 ground object pixels in total. These pixels contain a total of 16 types of ground objects.

Different proportions of samples for each dataset are used for training. For the IP dataset, 5%, 5%, and 90% of the labeled samples are randomly selected for training, validation, and testing, respectively. For the PC and WHU-HC datasets, 1%, 1%, and 98% of labeled samples are randomly selected for training, validation, and testing, respectively. For the HR-L dataset, 3%, 3%, and 94% of the labeled samples are randomly selected for training, validation, and testing, respectively. The details of dataset partitioning can be seen in Tables I–IV. Moreover, to alleviate the influence of insufficient training

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TABLE I

NUMBER OF TRAINING, VALIDATION, AND TESTING SAMPLES FOR THE IP DATASET

No.	Name	Train.	Val.	Test.
1	Alfalfa	2	3	41
2	Corn-notill	71	72	1285
3	Corn-mintill	42	41	747
4	Corn	12	12	213
5	Grass-pasture	24	24	435
6	Grass-trees	37	36	657
7	Grass-pasture-mowed	2	1	25
8	Hay-windrowed	24	24	430
9	Oats	1	1	18
10	Soybean-notill	48	49	875
11	Soybean-mintill	122	123	2210
12	Soybean-clean	29	30	534
13	Wheat	10	10	185
14	Woods	63	63	1139
15	Buildings-Grass-Trees-Drives	20	19	347
14	Stone-Steel-Towers	5	4	84
	Total	512	512	9225

TABLE II

NUMBER OF TRAINING, VALIDATION, AND TESTING SAMPLES FOR THE PC DATASET

No.	Name	Train.	Val.	Test.
1	Water	659	660	64652
2	Trees	76	76	7446
3	Asphalt	31	31	3028
4	Self-Blocking Bricks	27	27	2631
5	Bitumen	66	66	6452
6	Tiles	92	93	9063
7	Shadows	73	73	7141
8	Meadowsv	428	428	41970
9	Bare Soil	29	28	2806
	Total	1481	1482	145189

TABLE III

NUMBER OF TRAINING, VALIDATION, AND TESTING SAMPLES FOR THE HR-L DATASET

No.	Name	Train.	Val.	Test.
1	Dense Urban Fabric	8	9	271
2	Mineral Extraction Sites	2	2	63
3	Non Irrigated Arable Land	17	16	509
4	Fruit Trees	3	2	74
5	Olive Groves	42	42	1317
6	Broad-leaved Forest	6	7	210
7	Coniferous Forest	15	15	470
8	Mixed Forest	32	32	1008
9	Dense Sclerophyllous Vegetation	114	114	3565
10	Sparce Sclerophyllous Vegetation	84	84	2635
11	Sparcely Vegetated Areas	12	12	380
12	Rocks and Sand	14	15	458
13	Water	42	42	1309
14	Coastal Water	14	13	424
	Total	405	405	12693

data, the training samples are further enriched by several data augmentation methods, including vertical flip, horizontal flip, 90° rotation, 180° rotation, and 270° rotation.

B. Experimental Settings

In comparative experiments with other models, for fairness, the batch size and epoch are set to 64 and 300, respectively, to update the training parameters in all the frameworks. The optimizer and learning rate scheduler are set the same as the original to expect the performance of the other models in the original article. For our method, we used a stochastic gradient decent (SGD) optimizer with a momentum of 0.9 and a weight decay of 0.0001 to update the training parameters, while setting the learning rate to a constant 0.001. The NVIDIA GeForce RTX 3080 GPU with 10 GB of memory is used to accelerate the experiments. In our implementation, the parameters of the transformer architecture are set as follows. For the first MSFE module, the spectral dimension is mapped to 256 and then halved as the number of stages of the MSFE increases. The number of MHSA heads also varies between stages of MSFE, with the first stage set to 32 and then halved in turn. In the case of a network with three MSFEs, the spectral dimension for three stages is set as $\{256, 128, 64\}$, and the number of heads in the MHSA for different stages is $\{32, 16, 8\}$.

TABLE IV

NUMBER OF TRAINING, VALIDATION, AND TESTING SAMPLES FOR THE WHU-HC DATASET

No.	Name	Train.	Val.	Test.
1	Strawberry	447	448	43840
2	Cowpea	228	227	22298
3	Soybean	103	103	10081
4	Sorghum	54	53	5246
5	Water spinach	12	12	1176
6	Watermelon	45	46	4442
7	Greens	59	59	5785
8	Trees	179	180	17619
9	Grass	94	95	9280
10	Red roof	105	105	10306
11	Gray roof	169	169	16573
12	Plastic	37	37	3605
13	Bare soil	91	91	8934
14	Road	186	185	18189
15	Bright object	12	11	1113
16	Water	754	754	73893
	Total	2575	2575	252380

Overall classification accuracy (OA), average classification accuracy (AA), and kappa coefficient (Kappa) are selected to evaluate the final classification results. OA refers to the ratio of the number of correctly classified pixels to the number of all the pixels. AA corresponds to the sum average of the classification accuracy of each single category in all the categories, which reflects the overall situation of different categories and can measure the stability of different categories. Kappa coefficient is an indicator used for consistency tests and can also be used to measure the effect of classification, i.e., the so-called consistency is the consistency between the model prediction and actual classification results.

To reduce the effects of experimental randomness, we randomly split the labeled samples five times for five rounds of experiments and report their mean and standard deviation. Moreover, the classification color maps are visualized to qualitatively compare the results obtained by different methods.

C. Impact of the Input Spatial Size

The spatial input size determines how much spatial information around a pixel is used to classify that pixel. Therefore, the spatial input size plays an important role for HSI

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TABLE V
EFFECT OF DIFFERENT NUMBERS OF MSFE MODULES ON OA

Number	IP	PC	HR-L	WHU-HC
1	93.72	98.47	81.23	94.10
2	94.10	98.92	81.58	94.55
3	**94.92**	**99.30**	**82.16**	95.26
4	94.31	99.27	81.87	**95.30**
5	93.98	99.11	80.34	95.19

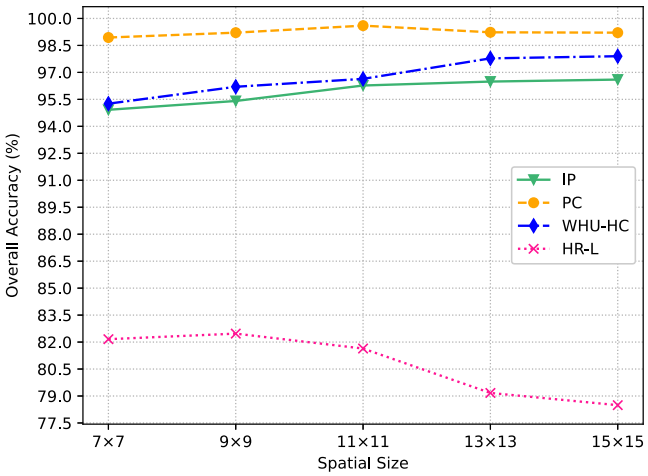


Fig. 5. OA (%) with different spatial sizes on the four datasets.

classification. In this part of the experiments, a set of spatial sizes $\{7, 9, 11, 13, 15\}$ are considered to explore their impact on the classification results, where the results are shown in Fig. 5. It can be seen that for the IP and WHU-HC datasets, the classification accuracy gradually increases as the spatial size increases. This is because these two datasets have relatively large smooth regions, and more pixels of the same type are used for classification as the spatial size increases. But the best results are achieved at 11×11 and 9×9 for the PC and HR-L datasets, respectively, where the various types of objects are relatively scattered. When the spatial size increases, it contains more redundant information, which has an adverse effect on the classification.

D. Impact of the Number of MSFE Modules

The number of MSFE modules determines the depth of network mining features and has a great influence on the performance of classification. To find the optimal number, a set of comparative experiments with MSFE numbers ranging into $\{1, 2, 3, 4, 5\}$ are set up. For the network with the number of MSFE modules as one, since the output feature has only one level, it is sent to the linear classifier as a global feature. When the number of MSFE modules is greater than or equal to two, the features of different levels are transformed into the same shape and inputted into CLFF to complete the aggregation of multilevel features. The experimental results are shown in Table V. On the IP, PC, and HRL datasets, the network achieves the best performance when the number of MSFE is three. On the WHU-HC dataset, the network containing four MSFEs achieves the best performance, with not much advantage over that is three. It can be concluded that the classification results are poor when the network is too deep or too shallow. This is because high-level semantic features cannot be extracted when the number of network layers is relatively shallow. However, as the number of network layers deepens, oversmoothing or overcompression will occur, resulting in a decrease in the degree of discrimination between different nodes and the feature expression of the nodes themselves. Therefore, three is the optimal number of MSFE in the experiments.

E. Comparison With State-of-the-Art Methods

In this section, to evaluate the performance of our proposed model, a series of experiments over the four datasets are conducted. Several state-of-the-art HSI classification methods are adopted for comparison, including a multiscale 3-D deep CNN (M3-D-DCNN) [57], a CNN-based 3-D DL method (HybridSN) [58], a deep feature fusion network (DFFN) [59], an RSSAN [43], an attention-based bidirectional long–short-term memory network (AB-LSTM) [45], a transformer-based backbone network (SpectralFormer) [48], and a group-aware hierarchical transformer network (GAHT) [56]. For fairness, the spatial size of the input in each model is set to 7×7. The rest of the parameters are consistent with the parameters in the original paper of each model to reproduce its best performance.

The experimental results are shown in Tables VI–IX. On the IP dataset, MCAL performs much better than other methods, where OA, AA, and KAPPA reach 94.92%, 90.84%, and 94.20%, respectively. Similar results can also be found in the WHU-HC dataset and MCAL leads other models in the three indicators of OA, AA, and Kappa. For the PC dataset, each model can achieve relatively good classification results, but MCAL still achieves the highest classification accuracy. On the most challenging HR-L dataset, MCAL achieves 1.53%−−10.18% higher than other models in OA and also outperforms other models in AA and Kappa. It can also be seen from the experimental data that for the CNN-based networks such as HybridSN and M3-D-DCNN, the results are poor on the IP dataset. The reason is that the IP dataset is smoother, where the pixels of the same type are relatively concentrated. For smaller convolution kernels, the receptive field is limited and cannot take into account all the pixels in the patch. For the HR-L dataset with relatively scattered pixels of the same type, the transformer-based network such as SpectralFormer has the worst classification results. Although SpectralFormer can handle global information well, its ability to extract local information is limited, resulting in the incorporation of interference from other categories of pixels during classification. GAHT also achieves excellent classification results on the four datasets, but still lower than those of MCAL. The reason is that although GAHT embedded 1×1 convolution in the transformer to extract local information, it does not consider the multiscale features in HSI space. MCAL combines the MSC with the transformer, which can take into account the global information and multiscale local information. The experimental results have also proved the effectiveness of the combination.

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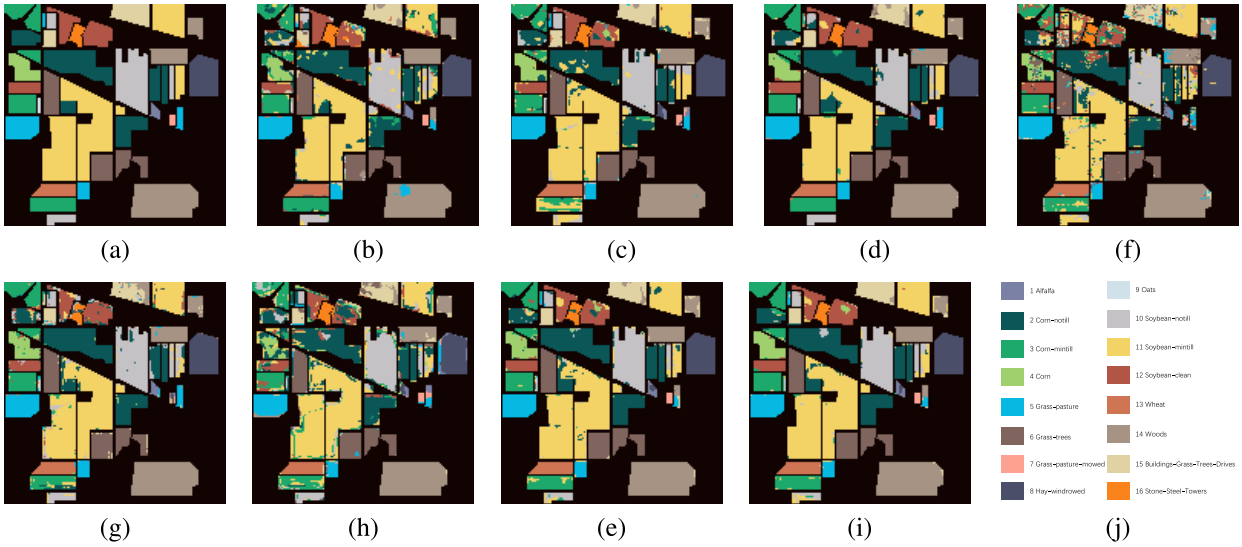


Fig. 6. Classification maps of the IP dataset with 5% training samples. (a) Ground truth. (b) HybridSN. (c) M3-D-DCNN. (d) DFFN. (e) AB-LSTM. (f) RSSAN. (g) SpectralFormer. (h) GAHT. (i) Proposed method. (j) Colors of objects.

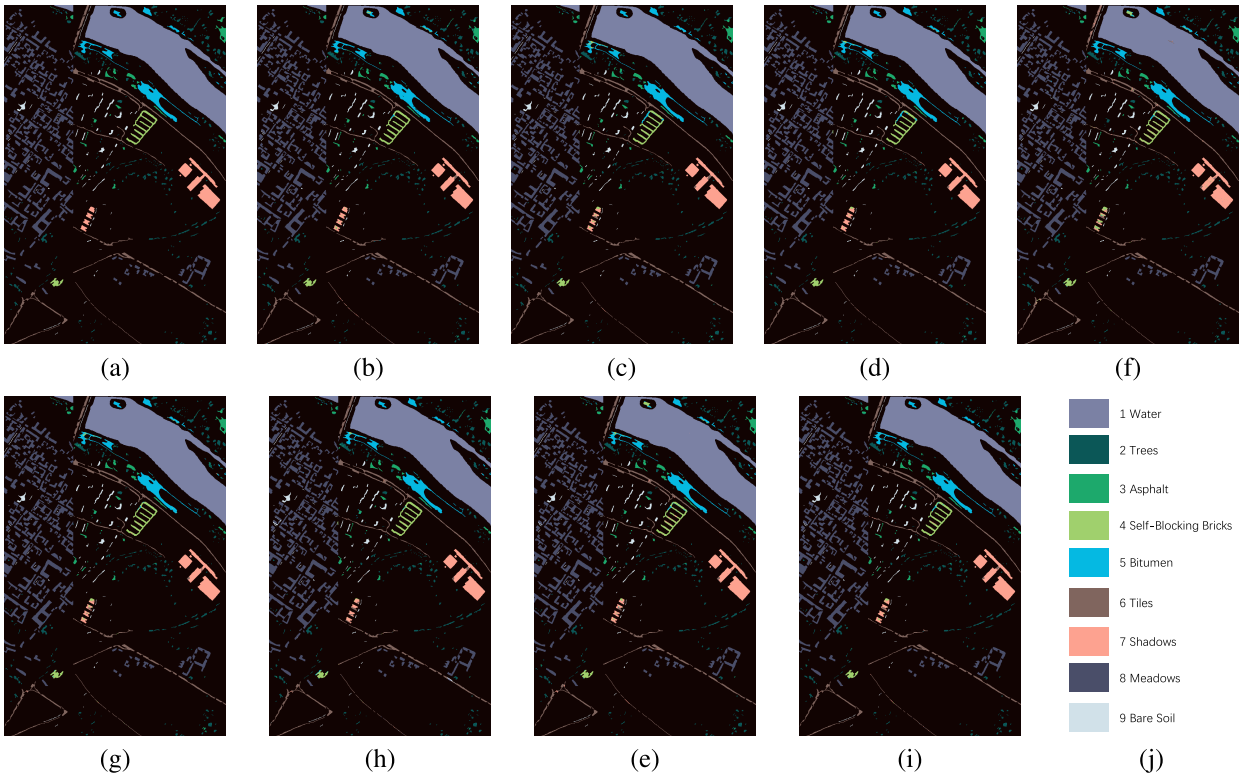


Fig. 7. Classification maps of the PC dataset with 1% training samples. (a) Ground truth. (b) HybridSN. (c) M3-D-DCNN. (d) DFFN. (e) AB-LSTM. (f) RSSAN. (g) SpectralFormer. (h) GAHT. (i) Proposed method. (j) Colors of objects.

We choose the one with the highest OA value among the five experiments to visualize the predicted samples of different methods. As shown in Figs. 6–9, on the four benchmark datasets, the proposed MCAL provides the results closest to the ground truth with less noise. This further illustrates its superiority.

F. Ablation Experiments

To analyze the contributions of different components in MCAL, the ablation experiments are constructed on four datasets. As described in Section III-A, our proposed network consists of three modules: SAM, MSFE, and CLFF module. Below, the importance of these three modules to MCAL will be verified through experiments, respectively.

1) Spectral Attention Module: In this session, the SAM is removed to train on four datasets. As shown in Table X, on the four datasets, the network performance without the SAM is inferior to MCAL. These results show that the SAM can indeed play a role in emphasizing useful bands and

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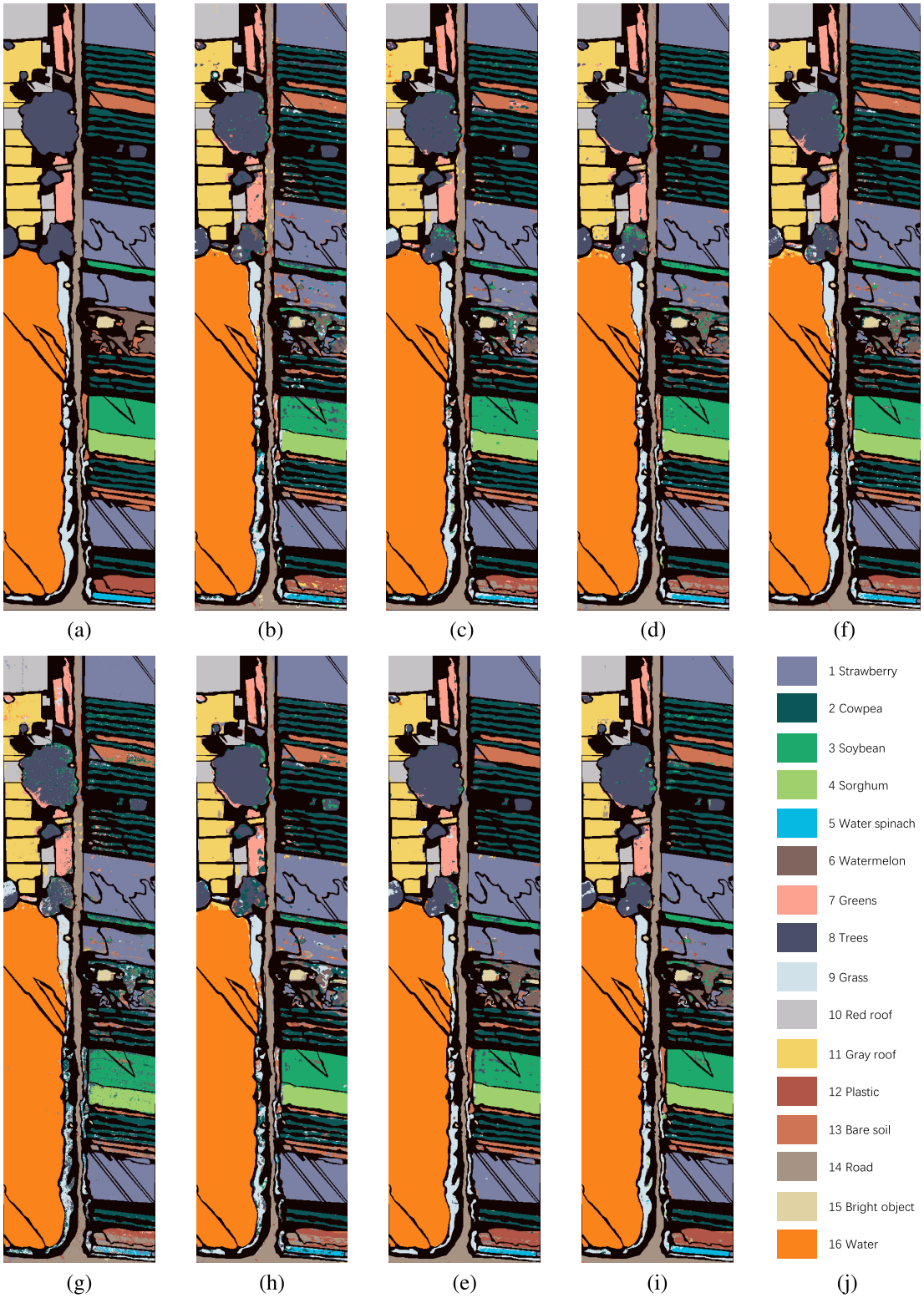


Fig. 8. Classification maps of the WHU-HC dataset with 1% training samples. (a) Ground truth. (b) HybridSN. (c) M3-D-DCNN. (d) DFFN. (e) AB-LSTM. (f) RSSAN. (g) SpectralFormer. (h) GAHT. (i) Proposed method. (j) Colors of objects.

suppressing useless bands. Adaptively recalibrating the spectral bands in this way facilitates spectral feature representation and final classification.

2) MSFE Module: This part of the experiments verifies the contribution of our proposed MSFE module to classification. Since it is the backbone in the network, it cannot be deleted directly. Therefore, the VIT module is chosen to replace it. As shown in Table X, MCAL using the MSFE module significantly outperforms the network using the VIT module on the four datasets. Such phenomenon indicates the effectiveness of our design for the combination of the convolutional layers and transformer structures, which can extract multiscale

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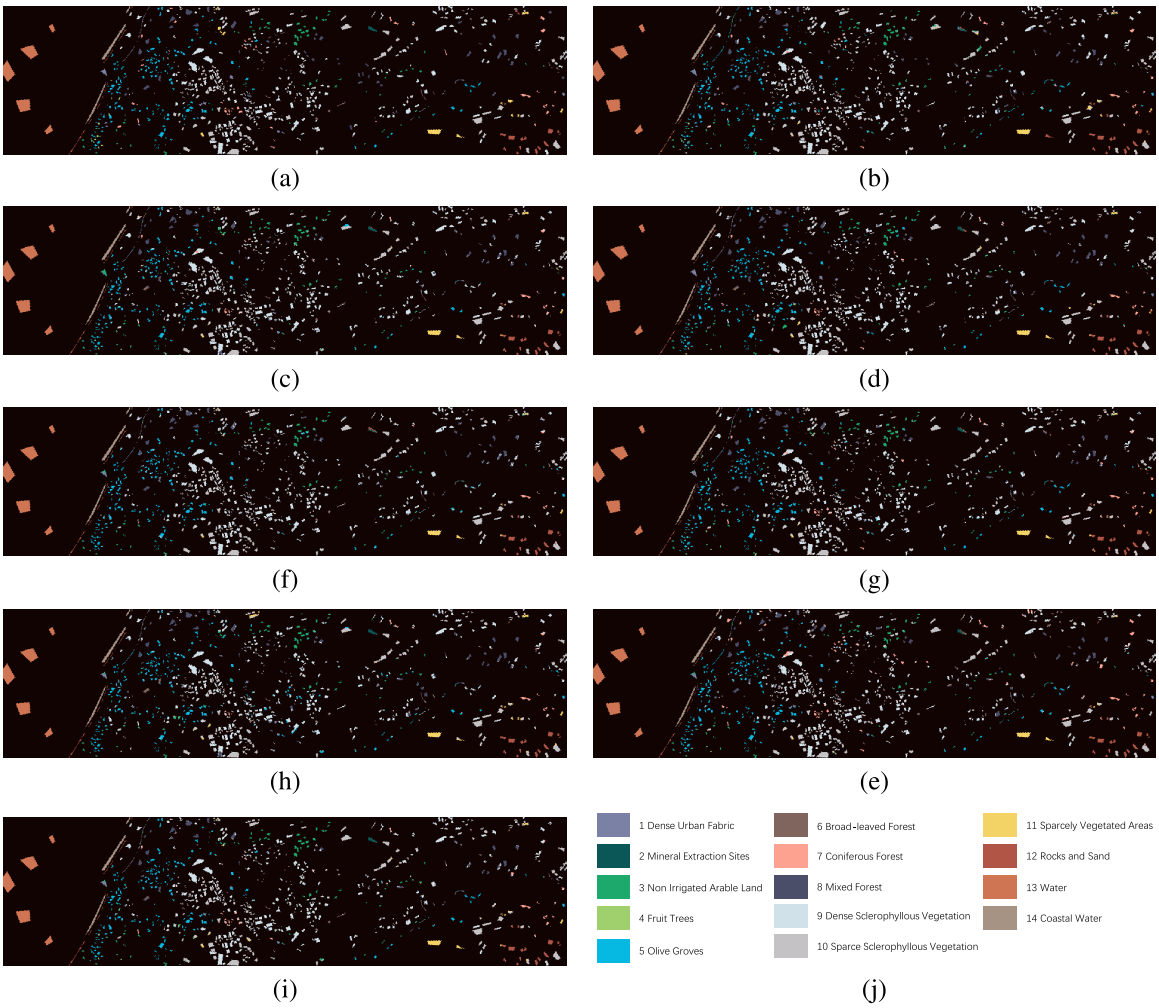


Fig. 9. Classification maps of the HR-L dataset with 3% training samples. (a) Ground truth. (b) HybridSN. (c) M3-D-DCNN. (d) DFFN. (e) AB-LSTM. (f) RSSAN. (g) SpectralFormer. (h) GAHT. (i) Proposed method. (j) Colors of objects.

TABLE VI

CLASSIFICATION RESULTS OF DIFFERENT CLASSIFICATION METHODS ON THE IP DATASET WITH 5% TRAINING SAMPLES

Class	M3D-DCNN [57]	HybridSN [58]	DFFN [59]	RSSAN [43]	AB-LSTM [45]	SpectralFormer [48]	GAHT [56]	Proposed
1	55.12 ± 21.18	**74.15 ± 12.11**	57.07 ± 22.44	32.20 ± 23.26	14.15 ± 27.09	24.39 ± 7.07	52.85 ± 24.09	67.32 ± 18.29
2	80.90 ± 1.22	75.94 ± 3.87	89.87 ± 4.92	88.14 ± 3.61	77.04 ± 4.07	67.89 ± 5.26	91.62 ± 2.90	**93.70 ± 2.93**
3	75.98 ± 7.39	81.18 ± 5.25	**96.65 ± 1.10**	83.59 ± 6.42	51.83 ± 25.71	64.58 ± 6.24	93.53 ± 1.20	95.15 ± 2.59
4	71.64 ± 11.52	57.75 ± 8.82	84.32 ± 7.46	72.58 ± 13.83	26.67 ± 21.79	49.01 ± 4.06	**89.36 ± 4.21**	89.30 ± 2.53
5	86.62 ± 4.44	84.97 ± 3.80	92.18 ± 2.93	89.79 ± 3.76	80.46 ± 6.82	74.30 ± 3.99	92.18 ± 3.26	**92.92 ± 3.10**
6	99.09 ± 0.75	96.35 ± 1.84	98.39 ± 0.76	97.20 ± 2.16	95.31 ± 1.63	91.60 ± 3.85	99.09 ± 0.65	**99.73 ± 0.28**
7	86.40 ± 16.32	77.60 ± 19.03	89.60 ± 18.86	56.80 ± 28.67	6.40 ± 12.80	43.20 ± 24.45	93.33 ± 9.43	**94.40 ± 9.33**
8	99.02 ± 0.89	97.81 ± 2.09	98.74 ± 1.48	98.00 ± 1.90	90.56 ± 7.75	93.95 ± 4.69	99.46 ± 0.77	**99.67 ± 0.65**
9	57.78 ± 18.12	32.22 ± 11.33	56.67 ± 24.44	21.11 ± 8.89	0.00 ± 0.00	31.11 ± 8.31	53.70 ± 13.09	**61.11 ± 25.82**
10	81.03 ± 6.89	77.71 ± 2.05	91.25 ± 2.74	83.15 ± 4.82	57.12 ± 20.64	75.18 ± 3.84	89.49 ± 2.08	**93.10 ± 2.88**
11	89.42 ± 2.59	84.94 ± 3.94	95.48 ± 0.29	91.87 ± 2.38	79.36 ± 4.4	81.44 ± 2.14	94.13 ± 0.73	**95.96 ± 0.89**
12	83.03 ± 2.84	73.93 ± 3.37	87.72 ± 4.19	76.33 ± 8.77	44.72 ± 18.93	47.87 ± 3.18	90.86 ± 4.14	**91.35 ± 3.95**
13	99.78 ± 0.26	92.43 ± 3.47	99.14 ± 1.00	96.76 ± 1.13	92.86 ± 3.69	97.08 ± 1.16	**99.82 ± 0.25**	99.79 ± 0.22
14	96.72 ± 1.70	96.08 ± 0.40	97.61 ± 1.33	97.03 ± 1.91	95.28 ± 2.16	92.98 ± 3.09	**97.92 ± 0.69**	97.38 ± 1.81
15	80.52 ± 7.78	72.10 ± 5.96	84.90 ± 5.30	75.91 ± 9.57	43.17 ± 13.20	69.91 ± 7.54	87.76 ± 5.03	**87.78 ± 5.30**
16	88.57 ± 6.84	81.43 ± 17.73	96.43 ± 2.92	86.19 ± 11.56	84.29 ± 13.69	81.43 ± 17.73	94.44 ± 4.05	**94.76 ± 5.03**
OA (%)	87.12 ± 0.86	83.69 ± 1.32	93.57 ± 1.16	88.66 ± 1.40	73.46 ± 6.05	76.86 ± 2.35	93.58 ± 0.20	**94.92 ± 0.57**
AA (%)	83.23 ± 1.78	78.54 ± 2.25	88.50 ± 2.41	77.92 ± 3.46	58.70 ± 7.70	68.84 ± 2.03	88.72 ± 1.55	**90.84 ± 2.34**
$\kappa\times100$	85.27 ± 0.96	81.38 ± 1.49	92.67 ± 1.32	87.04 ± 1.62	69.47 ± 7.23	73.59 ± 2.66	92.68 ± 0.24	**94.20 ± 0.65**

features while taking into account both the local and global information.

3) CLFF Module: For this experiment, the CLFF module is removed and the global features are directly used for classification. The experimental results are shown in Table X. On the four datasets, the effect of using global features for classification is not as good as that of using aggregated multilevel features. This is because the global features only

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TABLE VII

CLASSIFICATION RESULTS OF DIFFERENT CLASSIFICATION METHODS ON THE PC DATASET WITH 1% TRAINING SAMPLES

Class	M3D-DCNN [57]	HybridSN [58]	DFFN [59]	RSSAN [43]	AB-LSTM [45]	SpectralFormer [48]	GAHT [56]	Proposed
1	99.97 ± 0.05	99.98 ± 0.03	99.99 ± 0.02	99.99 ± 0.02	99.97 ± 0.05	99.99 ± 0.02	99.97 ± 0.02	**100.00 ± 0.00**
2	96.03 ± 1.64	95.41 ± 2.90	95.82 ± 0.56	96.47 ± 0.75	93.83 ± 2.48	95.54 ± 1.39	96.97 ± 0.61	**96.74 ± 1.74**
3	93.71 ± 1.28	93.17 ± 4.02	93.56 ± 3.05	93.17 ± 4.02	92.02 ± 2.21	92.54 ± 3.42	91.46 ± 2.98	**93.77 ± 1.04**
4	90.91 ± 3.93	**94.51 ± 2.47**	93.38 ± 1.75	93.71 ± 2.77	77.53 ± 8.24	89.32 ± 5.99	94.44 ± 2.25	91.87 ± 4.04
5	94.07 ± 1.66	97.17 ± 0.95	97.06 ± 1.38	96.03 ± 1.85	94.60 ± 1.77	96.41 ± 1.56	96.97 ± 2.30	**98.89 ± 0.39**
6	99.34 ± 0.26	98.27 ± 0.34	99.01 ± 0.56	98.93 ± 0.18	96.50 ± 1.03	95.97 ± 1.10	98.69 ± 0.24	**99.38 ± 0.32**
7	92.70 ± 0.92	96.02 ± 0.77	93.16 ± 2.46	94.17 ± 0.90	90.14 ± 1.28	93.70 ± 0.90	95.11 ± 1.39	**97.05 ± 0.92**
8	99.91 ± 0.04	99.88 ± 0.06	99.88 ± 0.06	99.84 ± 0.08	99.53 ± 0.26	99.81 ± 0.09	**99.96 ± 0.01**	99.94 ± 0.04
9	99.59 ± 0.64	98.80 ± 0.87	99.62 ± 0.25	99.25 ± 0.45	**99.92 ± 0.03**	96.05 ± 1.24	99.88 ± 0.07	99.81 ± 0.20
OA (%)	98.79 ± 0.11	99.03 ± 0.12	98.96 ± 0.18	98.94 ± 0.08	98.03 ± 0.21	98.57 ± 0.20	99.08 ± 0.08	**99.30 ± 0.07**
AA (%)	96.25 ± 0.47	97.02 ± 0.40	96.83 ± 0.48	96.71 ± 0.40	93.84 ± 0.74	95.50 ± 0.90	97.05 ± 0.22	**97.50 ± 0.43**
$\kappa\times100$	98.28 ± 0.16	98.62 ± 0.17	98.52 ± 0.25	98.50 ± 0.11	97.21 ± 0.30	97.97 ± 0.29	98.70 ± 0.11	**99.01 ± 0.11**

TABLE VIII

CLASSIFICATION RESULTS OF DIFFERENT CLASSIFICATION METHODS ON THE HR-L DATASET WITH 3% TRAINING SAMPLES

Class	M3D-DCNN [57]	HybridSN [58]	DFFN [59]	RSSAN [43]	AB-LSTM [45]	SpectralFormer [48]	GAHT [56]	Proposed
1	54.39 ± 8.49	39.93 ± 16.96	37.71 ± 17.86	49.37 ± 15.29	35.13 ± 20.10	47.75 ± 9.05	48.91 ± 2.25	**55.35 ± 6.08**
2	90.79 ± 12.55	80.00 ± 14.64	79.68 ± 16.35	61.27 ± 26.00	70.48 ± 11.47	**98.41 ± 3.17**	86.98 ± 15.11	66.98 ± 25.46
3	82.28 ± 3.25	70.37 ± 6.81	80.16 ± 4.35	82.08 ± 2.57	80.04 ± 7.59	66.44 ± 6.55	**88.35 ± 5.50**	88.06 ± 6.28
4	**30.54 ± 11.36**	17.57 ± 17.66	4.86 ± 7.33	14.59 ± 9.10	0.00 ± 0.00	1.89 ± 2.36	15.11 ± 5.14	25.68 ± 13.99
5	85.92 ± 3.12	83.40 ± 3.19	88.93 ± 3.48	90.33 ± 4.14	79.65 ± 6.23	70.90 ± 4.22	**93.69 ± 1.76**	90.98 ± 3.55
6	48.67 ± 10.67	20.10 ± 4.18	27.52 ± 6.41	34.48 ± 5.81	10.95 ± 15.84	13.05 ± 7.07	**58.96 ± 14.03**	48.00 ± 10.95
7	**64.55 ± 8.41**	35.19 ± 18.69	46.98 ± 9.40	56.77 ± 5.95	45.96 ± 9.57	42.21 ± 5.77	62.60 ± 14.00	58.43 ± 8.38
8	68.61 ± 8.23	56.53 ± 20.89	61.75 ± 7.19	70.54 ± 5.84	62.62 ± 4.13	62.28 ± 6.11	65.23 ± 4.85	**70.62 ± 5.40**
9	78.91 ± 3.89	81.01 ± 6.27	82.12 ± 2.17	79.72 ± 2.14	82.51 ± 2.28	72.07 ± 4.82	78.38 ± 5.22	**82.74 ± 2.64**
10	**81.98 ± 3.66**	73.46 ± 5.86	78.68 ± 4.04	80.88 ± 4.24	81.24 ± 2.48	71.43 ± 5.95	81.71 ± 1.06	80.52 ± 2.34
11	60.42 ± 6.80	52.58 ± 7.18	56.89 ± 5.59	59.42 ± 7.69	59.74 ± 7.46	57.11 ± 5.50	62.41 ± 5.64	**67.26 ± 6.06**
12	92.58 ± 4.03	84.19 ± 10.97	90.13 ± 2.82	86.29 ± 5.35	90.66 ± 3.44	91.09 ± 2.89	91.72 ± 1.06	**93.67 ± 1.04**
13	**100.00 ± 0.00**	**100.00 ± 0.00**	**100.00 ± 0.00**	**100.00 ± 0.00**	**100.00 ± 0.00**	**100.00 ± 0.00**	**100.00 ± 0.00**	**100.00 ± 0.00**
14	**100.00 ± 0.00**	99.62 ± 0.75	**100.00 ± 0.00**	99.91 ± 0.12	99.91 ± 0.12	99.95 ± 0.09	**100.00 ± 0.00**	99.95 ± 0.09
OA (%)	80.63 ± 1.15	75.21 ± 1.88	78.78 ± 0.87	80.10 ± 0.39	78.18 ± 1.32	71.98 ± 0.98	81.11 ± 1.29	**82.16 ± 0.34**
AA (%)	74.26 ± 1.15	63.85 ± 4.25	66.82 ± 2.08	68.97 ± 1.77	64.21 ± 1.65	63.90 ± 0.47	73.86 ± 2.41	**74.82 ± 2.16**
$\kappa\times100$	76.96 ± 1.25	70.28 ± 2.54	74.54 ± 1.06	76.26 ± 0.44	73.81 ± 1.54	66.54 ± 1.06	77.53 ± 1.45	**78.72 ± 0.42**

TABLE IX

CLASSIFICATION RESULTS OF DIFFERENT CLASSIFICATION METHODS ON THE WHU-HC DATASET WITH 1% TRAINING SAMPLES

Class	M3D-DCNN [57]	HybridSN [58]	DFFN [59]	RSSAN [43]	AB-LSTM [45]	SpectralFormer [48]	GAHT [56]	Proposed
1	96.67 ± 0.36	95.47 ± 0.83	97.29 ± 0.68	97.32 ± 0.65	95.43 ± 0.89	96.30 ± 0.78	97.73 ± 0.12	**98.23 ± 0.42**
2	91.07 ± 2.40	86.87 ± 1.38	95.07 ± 1.55	94.65 ± 1.14	86.55 ± 2.11	84.95 ± 2.43	93.24 ± 1.09	**95.36 ± 0.62**
3	90.51 ± 1.92	84.73 ± 4.15	95.42± 1.26	92.66 ± 2.74	79.27 ± 1.74	91.40 ± 1.62	**95.54 ± 2.03**	94.41 ± 2.34
4	96.23 ± 1.21	96.43 ± 1.04	96.54 ± 1.07	96.30 ± 0.59	90.98 ± 1.46	95.33 ± 0.67	97.21 ± 1.06	**97.35 ± 0.99**
5	62.04 ± 11.86	61.96 ± 10.46	**86.80 ± 7.24**	82.16 ± 5.97	35.73 ± 13.94	66.19 ± 9.42	86.60 ± 4.18	86.33 ± 7.89
6	53.81 ± 5.83	54.06 ± 7.24	64.98 ± 7.25	61.30 ± 4.65	20.32 ± 6.49	41.58 ± 3.59	**72.99 ± 4.22**	68.93 ± 2.20
7	84.12 ± 5.07	87.07 ± 2.38	89.38 ± 2.63	87.11 ± 6.90	77.20 ± 2.07	80.03 ± 2.37	88.70 ± 1.39	**90.08 ± 2.42**
8	82.53 ± 3.82	82.98 ± 3.41	88.61 ± 0.11	89.00 ± 2.06	72.62 ± 3.44	77.98 ± 2.19	**89.52 ± 2.91**	89.38 ± 1.47
9	83.85 ± 2.49	81.09 ± 2.98	91.23 ± 1.49	88.04 ± 2.31	69.05 ± 4.48	71.73 ± 3.68	92.52 ± 1.20	**92.63 ± 2.46**
10	95.77 ± 0.71	97.54 ± 0.48	98.47 ± 0.37	96.86 ± 0.78	94.58 ± 1.01	96.65 ± 0.44	**98.78 ± 0.33**	98.13 ± 0.62
11	95.22 ± 1.00	94.71 ± 1.78	95.81 ± 1.53	**96.96 ± 1.51**	92.89 ± 2.42	95.07 ± 1.35	95.61 ± 0.70	96.18 ± 1.28
12	70.37 ± 6.07	62.81 ± 9.11	78.80 ± 7.46	74.19 ± 8.30	38.04 ± 11.76	55.23 ± 3.87	82.84 ± 1.59	**83.42 ± 3.82**
13	70.85 ± 2.04	70.63 ± 4.48	**79.95 ± 2.16**	75.37 ± 3.76	62.21 ± 3.00	67.34 ± 2.91	79.65 ± 3.15	79.08 ± 2.24
14	89.72 ± 1.90	88.22 ± 2.32	91.93 ± 1.88	92.11 ± 1.08	86.38 ± 0.91	87.90 ± 1.41	**95.54 ± 0.32**	94.34 ± 1.29
15	81.53 ± 3.60	55.33 ± 7.15	79.30 ± 6.66	74.68 ± 7.51	71.48 ± 5.46	62.26 ± 12.18	81.80 ± 5.45	**84.96 ± 4.70**
16	99.30 ± 0.26	99.47 ± 0.14	99.30 ± 0.22	99.49 ± 0.22	99.04 ± 0.58	96.90 ± 2.21	99.55 ± 0.07	**99.59 ± 0.17**
OA (%)	92.04 ± 0.41	90.98 ± 0.25	94.57 ± 0.70	94.03 ± 0.44	87.43 ± 0.60	89.92 ± 0.30	95.15 ± 0.12	**95.26 ± 0.22**
AA (%)	83.97 ± 0.93	81.21 ± 0.72	89.30 ± 1.91	87.39 ± 1.39	73.24 ± 1.65	79.32 ± 0.78	90.49 ± 0.20	**90.52 ± 0.83**
$\kappa\times100$	90.67 ± 0.48	89.45 ± 0.29	93.65 ± 0.81	93.00 ± 0.52	85.26 ± 0.71	88.07 ± 0.35	94.32 ± 0.29	**94.44 ± 0.26**

consider the deep-level features, which may reduce the degree of discrimination between different nodes due to overcompression. Aggregated features can avoid the problems caused by overcompression by fusing features at different levels and take into account the correlation of features at various levels. Therefore, aggregated features are more suitable for the HSI classification tasks than global features.

G. Robustness Evaluation

To evaluate the robustness of the proposed network, our method is tested with other competitors along with the increasing number of training samples. Specifically, 5%, 10%, 15%, and 20% labeled samples were randomly chosen as the training data for the IP dataset, while 2%, 3%, 4%, and 5% for the HR-L dataset, and 0.25%, 0.5%, 1%, and 2% labeled samples

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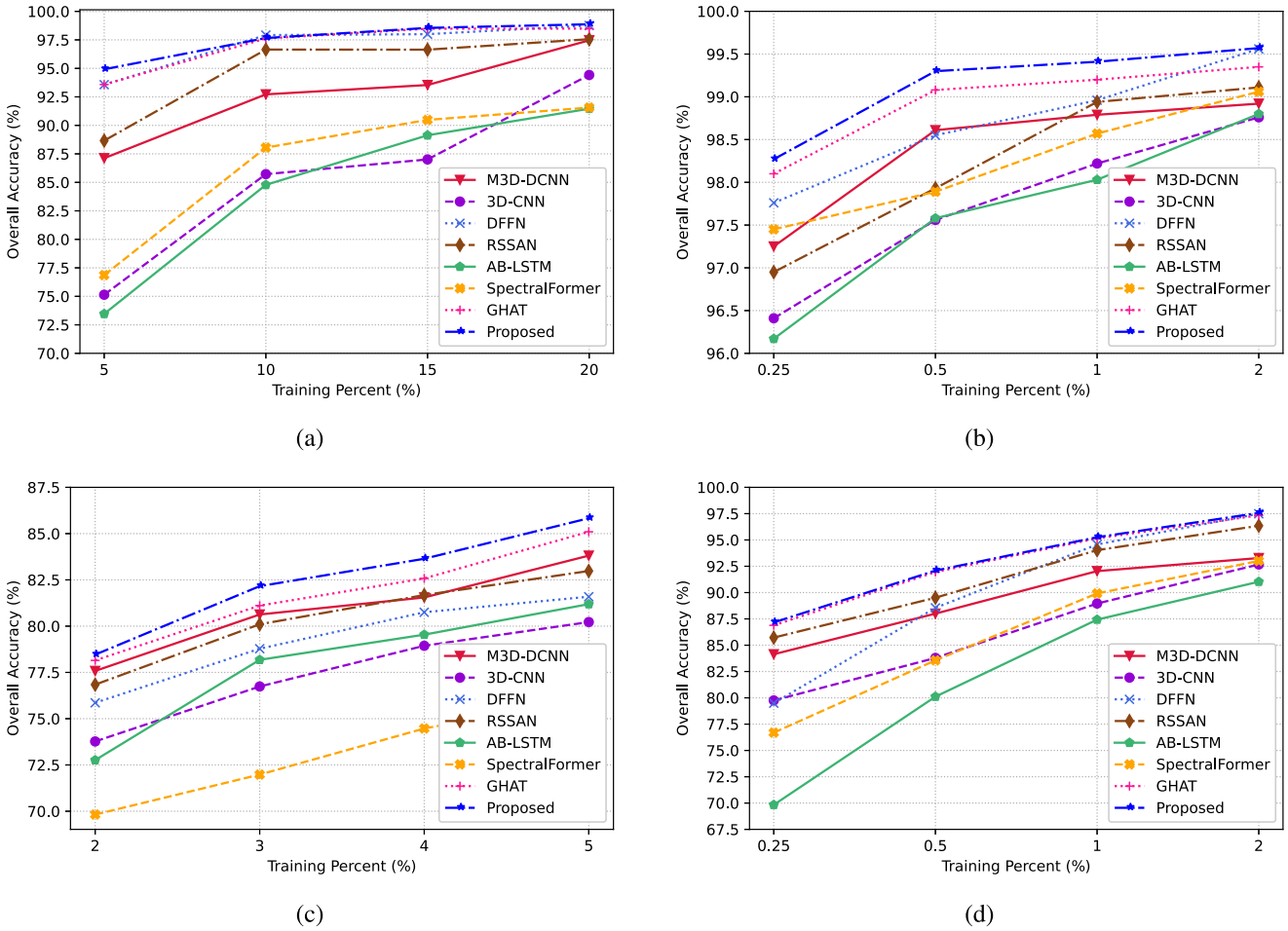


Fig. 10. Classification results versus different percentages of training samples on the four datasets. (a) IP. (b) PC. (c) HR-L. (d) WHU-HC.

TABLE X
EFFECT OF DIFFERENT COMPONENTS ON OA

Dataset	IP	PC	HR-L	WHU-HC
MSFE+CLFF	94.68	98.91	81.67	94.59
SAM+VIT+CLFF	93.78	98.28	81.06	94.37
SAM+MSFE	94.01	98.47	81.35	94.24
SAM+MSFE+CLFF	**94.92**	**99.30**	**82.16**	**95.26**

were randomly chosen as the training data for the PC and WHU-HC datasets.

Fig. 10 shows the results of the overall accuracy (OA) of different classifiers. It can observed that the proposed method significantly outperforms other methods in the majority of cases and is only close to DFFN on the IP dataset and GAHT on the WHU-HC dataset. Especially when the number of training samples is limited, our proposed method can still achieve high classification accuracy.

V. CONCLUSION

This article proposes a novel HSI classification method called MCAL, which includes three main modules: SAM, MSFE, and CLFF. First, the SAM calibrates the spectral bands, aiming to enhance the useful information in the spectrum and suppress unwanted noise. Then, the MSFE is constructed to explore both the multiscale local spatial features of pixels using different sizes of convolutions and global features of pixels using MHSA. The extracted features take into account both the global information and local information, making up for the defect of the transformer in capturing local features. Next, in the CLFF, the low-, middle-, and high-level features extracted from MSFE are integrated in an attentional manner to deeply mine multilevel features and global features in HSI. Finally, the extracted depth features are linearized into a 1-D vector to complete the classification. A series of experiments over four benchmark datasets demonstrate that the proposed MCAL obviously outperforms several state-of-the-art networks for hyperspectral classification.

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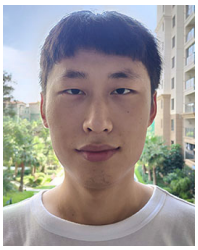
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